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Title of the article: *An ultrasensitive photoelectrochemical platform for quantifying photoinduced electron-transfer properties of a single entity*

The content extracted is as follows:

Among the most efficient photoelectrochemical reactions are those involving light absorption by wide-band-gap semiconductor nanoparticles and molecular adsorbates, which result in the occurrence of electron-transfer reactions at the semiconductor interface that are the basis for photoelectric conversion in photovoltaic devices^{2,8,9}.

However, the key bottleneck in applying single-entity electrochemistry to photoelectrochemical measurement is the need for realtime recording of the ultrafast photoinduced electron-transfer data³⁶.

That is, the electron-transfer process of a photoelectrochemical reaction occurs on a sub-picosecond timescale and cannot be electrochemically resolved because of the limitations of current electrochemical instruments.

Current photoinduced electron-transfer research is mainly based on ultrafast time-resolved transient spectroscopy techniques for tracking the intermediate states of the electron-transfer process in photoelectrochemical reactions on picosecond and femtosecond timescales³⁷⁻³⁹.

Therefore, in our photoelectrochemical measurements, the rate-determining steps are the collision of a nanoparticle with the TiO₂@Au UME and/or the electron diffusion process inside the TiO₂ lm.

To elucidate the electron-transfer dynamics of the collision process, we studied the photoelectrochemical behaviors of individual dye-tagged oxide semiconductor nanoparticles colliding with a bare Au UME. No current signal was observed under illumination in this case (Supplementary Fig. 1).

In this study, we performed single-entity photoelectrochemical experiments using a low-noise electrochemical instrument at a sampling rate of 100 kHz.

As a result, the typical duration of electron transit in TiO₂ lm was considerably increased, allowing the achievement of successful real-time detection of single-entity photoelectrochemical behaviors.

Owing to a trap-mediated electron diffusion process in TiO₂ lms, we observed submillisecond photocurrent transients associated with single-entity photoelectrochemical reactions.

Based on high-resolution photoelectrochemical measurements, our platform can be used for quantifying the photoinduced electron-transfer process of a single event, enabling electron-transfer dynamics analysis at a semiconducting TiO₂ lm.

It will certainly help to achieve an improved understanding of the electron-transfer process at the semiconductor interface, where the photoelectrochemical reaction processes take place.

In addition, considering the extreme sensitivity of individual event signals, testing the effects of the filter and amplifier of a low-noise electrochemical instrument should be performed as a best practice standard in single-entity photoelectrochemical studies to minimize the electronics effects on measurement results^{48,49}.

Higher light intensity causes an increase in current background noise due to local thermal drift. In this protocol, we describe step-by-step procedures for the fabrication of TiO₂@Au UMEs, the preparation of monodispersed dye-tagged oxide semiconductor nanoparticle suspensions and the subsequent procedures to monitor the photoelectrochemical behaviors of individual entities in real time, process data and confirm single-nanoparticle events.

CRITICAL STEP The parameters used are highly dependent on the SEM instrument. ? TROUBLESHOOTING

In this protocol, we have described detailed procedures for fabrication of an ultrasensitive photoelectrochemical platform, which could be widely used for real-time monitoring of the electrontransfer process of a single entity in various photoelectrochemical systems.

Considering the electron recombination in a multiple-trapping TiO₂ lm, the effective number of collected electrons for a single N719 molecule per typical collision (N_e) at a TiO₂@Au UME from the measured transients was calculated by $N_e = Q/neN_{N719}$, where Q is the integrated charge of individual events, n is the number of electrons involved in the photoelectrochemical reaction (1 for N719), e is the elementary charge and N_{N719} is the average number of N719 molecules loaded onto the surface of a single TiO₂ nanoparticle surface (~600; Supplementary Fig. 9).

In addition, the measured results of TiO₂ lm thickness-dependent current traces further demonstrated that a trap-limited electron diffusion process was the rate-limiting step in our

singleentity photoelectrochemical measurement.